1 Related work

Universal and operator induction. Solomonoff induction predicts sequences with a universal prior over programs (Solomonoff, 1964, 1978; Li and Vitányi, 2008). Operator induction predicts the answer to a new question from an unordered set of question–answer pairs (Solomonoff, 1999, 2008); the companion papers develop its one-tape universal law and its versions (Hänni and Claude (Anthropic), 2026c,d), and **??** relates SAI to them version by version: SAI is operator induction in which the questions are sentences and deduction ties the answers together. Whether a universal prior justifies a preference for simplicity is disputed (Sterkenburg, 2016; on universal learning machines, Sterkenburg, 2019); nothing here depends on it. The computable, finite-class precursors are two earlier reports: Hänni and Claude (Anthropic) (2026a) study a Bayesian axiom inducer with template priors and derivation likelihoods, and coin-free axiom induction under time penalties, which **??** extends; Hänni and Claude (Anthropic) (2026b) learn axioms and schemas from their instances.

Probabilities on sentences. Logical probabilities on the sentences of a language go back to Carnap (1950); coherent probabilities on first-order sentences were studied by Gaifman (1964), Scott and Krauss (1966) and Gaifman and Snir (1982); Hutter et al. (2013) extend given probabilities of some sentences to all sentences of a higher-order logic under desiderata that include the Gaifman condition; Christiano (2014) gives a tractable algorithm; and logical inductors (Garrabrant et al., 2016a) assign probabilities to the sentences of a given theory and refine them over time. These are probabilities of truth, and a logical inductor is given the axioms, while SAI is given labelled sentences and is uncertain about the axioms. Bel, the probability that a random theory proves a sentence, is a belief function (Dempster, 1967; Shafer, 1976) with an independent share, and it satisfies the Gaifman condition exactly when the instances prove the universal sentence (**??**). Sawin and Demski (2013) show that computable distributions that come to believe the true Π_1 sentences must give some short true Π_2 sentences arbitrarily low limiting probability; **????** show a parallel divide for the ideal SAI posterior, not a consequence of theirs. Demski's prior (Demski, 2012), as it is usually described, draws random sentences and keeps each one consistent with those kept; the modification due to Fallenstein, Demski and Garrabrant (Garrabrant, 2015), used in Garrabrant et al. (2016b), draws theories as the outputs of a universal machine on random bits. So random axiom systems drawn by a universal machine are not new here: the sampling stage of M_A is that of the modified prior. Those processes iterate draws until the theory is complete; M_A uses a single draw, whose theory is incomplete with positive probability, and for complete theories proof-requiring and consistency-requiring compatibility coincide (**????**).

Learning theories from data. Formal learning theory studies identification in the limit (Gold, 1967; Putnam, 1965b), also of first-order theories (Osherson et al., 1991; Kelly, 1996; Martin and Osherson, 1998); **??** is a Bayesian form of Gold's distinction between text and informant. Shapiro's model inference, which infers theories from facts by conjectures and refutations (Shapiro, 1981, 1991), is the nearest learning-theoretic precedent of proof-requiring compatibility; it outputs one conjecture at a time, without a prior. Inductive logic programming (cf. Muggleton and De Raedt, 1994; Cropper and Dumančić, 2022) asks a hypothesis to entail the positive examples and to be consistent with the negative ones, a mixture of the two compatibility notions of **??**, invents predicates (Muggleton and Buntine, 1988), and justifies theories by compression (Muggleton, 1995; Conklin and Witten, 1994). In cognitive science, intuitive theories are sets of logical laws generated by a probabilistic grammar and learned by Bayesian inference with stochastic search

(Kemp et al., 2008, 2010; Ullman et al., 2012): models over finite spaces of hypotheses, of the same design as the template prior of Hänni and Claude (Anthropic) (2026a), where SAI is universal.

Random axioms and Gödel phenomena. What a probabilistic machine produces with positive probability can be produced deterministically (cf. de Leeuw et al., 1956); the lottery (**??**) is where this averaging fails for random theories, since majority votes over consistency constraints of unbounded arity can be inconsistent (cf. the lottery paradox, Kyburg, 1961). Its closer relative is that randomness helps for mass problems: a class of positive measure, such as a class of bounded diagonally non-computable functions, can have no computable member (cf. Kučera, 1985; Bienvenu and Porter, 2016). That randomisation can succeed where determinism cannot, below a success-probability threshold, is familiar for learners (Pitt, 1989; Ambainis, 1999); here the random objects are the hypotheses themselves. Bienvenu et al. (2014) add randomly chosen true complexity axioms to PA and show that they do not help to prove new interesting theorems but can shorten proofs unless NP = PSPACE; the threshold theories of **??** use a soundness threshold of a related kind, placed just above the unknown inconsistency probability of the universal random theory. **??** also meets Chaitin's theorem (Chaitin, 1974; Raatikainen, 1998), flexible formulas (cf. Kripke, 1962; Woodin, 2011), Kreisel's conjecture (cf. Hrubeš, 2009) and the problem of old evidence (Garber, 1983).

Craig, Ramsey, Hempel. Craig's theorem (Craig, 1953, 1956) is the background of Hempel's theoretician's dilemma (Hempel, 1958) and of its replies: Craigian axiomatisations are unmanageable; theoretical terms serve inductive systematisation (cf. Niiniluoto, 1972); science does not aim only at prediction (cf. Putnam, 1965a). **??** sets them against our results. In monadic inductive logic theoretical concepts can change inductive probabilities (cf. Niiniluoto and Tuomela, 1973); in the universal mixture they cannot, beyond a constant factor (**??**). Every r.e. theory has a polynomial-time axiomatisation by a variant of Craig's trick (Krajíček, 2025), and every r.e. theory without finite models has a finitely axiomatised conservative extension with one more binary relation symbol (Kleene, 1952; Craig and Vaught, 1958, as reported by Pakhomov and Visser, 2019); **??** is an effective many-sorted form. Ramsey sentences and Lewis's definitions of theoretical terms (Ramsey, 1931; Lewis, 1970) appear in **??**. The complexity facts we use are standard (cf. Fischer and Rabin, 1974; Cook and Reckhow, 1979; Chandra et al., 1981; on lengths of proofs, Gödel, 1936; Pudlák, 1998).

Usable theories. The size of a theory that a weak reasoner can use is studied in knowledge compilation (Selman and Kautz, 1996; Darwiche and Marquis, 2002), in compilation for complete unit resolution and propagation (del Val, 1994; Babka et al., 2013), and for encodings of XOR constraints (Gwynne and Kullmann, 2014), with monotone-circuit methods (Bessiere et al., 2009; Göös et al., 2019). The declarative/procedural question has a history in AI: the advice taker (McCarthy, 1959), the controversy discussed by Winograd (1975), and "algorithm = logic + control" (Kowalski, 1979). Khardon and Roth (1997) give learning-to-reason algorithms for a propositional class without efficient reasoning from a formula-based knowledge base, and program synthesis can learn a library of reusable abstractions (Ellis et al., 2021).

Laws, approximation, explanation, and what a theory is. Lewis's best-system account (Lewis, 1983, 1994) and its language-relative version (Cohen and Callender, 2009) are the counterpart of **??**, by analogy only. That a model includes its noise distribution, and that the precision of measurement enters the code length, are standard in MDL and MML (Rissanen, 1978; Wallace,

2005; Grünwald, 2007); the split into model and noise is the subject of algorithmic statistics (Gács et al., 2001; Vereshchagin and Vitányi, 2004); Akaike-based accounts give a statistical form of the dependence of selection on precision (Forster and Sober, 1994). Truthlikeness (cf. Miller, 1974; Niiniluoto, 1987) is a global notion where ours is local. The accounts of explanation (Hempel and Oppenheim, 1948; Kitcher, 1989; van Fraassen, 1980; Woodward, 2003) and of what a theory is (Suppe, 1977; Suppes, 1960; Sneed, 1971; Halvorson, 2012; Cartwright, 1983) are the background of **??**. Discovery systems find laws in data: BACON rediscovers Kepler's third law (Langley et al., 1987), symbolic regression finds physical laws (Schmidt and Lipson, 2009; Udrescu and Tegmark, 2020), and the Apperception Engine constructs symbolic causal theories of sensory sequences (Evans et al., 2021). They search, where SAI evaluates.

What is new. To our knowledge the following are new: the single-draw proof-requiring law M_A and its basic theory (**??**); the one-tape and Kolmogorov equivalences with consistent SOI and the transfer principle (**??????**), while the coin-free weight equivalence is Craig's construction inside the universal prior, proved for deciders in Hänni and Claude (Anthropic) (2026a); the lottery, its comparison of versions and its arithmetic versions with true data (**????**); the threshold theories on Π_1 data and the coding theorem for an inductive inference rule (**????**); the reasoner-relative separations inside a universal inducer (**??**), where the bound for systems of laws (**??**) is a short combination of known results that we did not find stated and that is probably known to specialists; a complexity-theoretic form of the impracticality reply to Hempel, which we did not find in the literature, and the derivation-bounded constructions and fit classes of **??**; and, for a proof-requiring inducer, the choice between honest laws, laws with exceptions and laws with error models (**????**). Not new are random axiom systems drawn by a universal machine, the mixture and dominance arguments, Craig's construction, the rules of conditioning for belief functions, and the reading of a law with its noise distribution as a two-part code.

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